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Detailed Course

Introduction and Motivation

General Context

The **k-means** algorithm is a method for **estimating discrete latent factors** in an **unsupervised** clustering context.

Course Objectives:

- Understand the **geometric and statistical** properties of data
- Analyze data and develop tools for this analysis
- Specifically address the **(unsupervised)** approach to data clustering
- Understand the **assumptions** made in designing these tools
- Work on the **theory in depth**

Important Note: Clustering is similar to classification (unsupervised), density estimation, and is dual to outlier detection

Real Data Problem

Fundamental Observation:

Data generally does **not** come from a simple Gaussian process (variations around a mean prototype)

Characteristics of Real Data:

- Data distribution typically shows **non-uniformity**
- Presence of **regions of higher density**

Definition of Clustering:

Clustering is an **unsupervised** method that aims to discover **coherent groups** of data, corresponding to **density peaks** in the data

Alternative Terminology

Vector Quantization (VQ):

Often used synonym (in signal processing) for clustering, meaning "multi-dimensional quantization"

Overview of Clustering Methods

Main Methods

There are many clustering methods, including:

Hierarchical Methods:

- Hierarchical Agglomerative Clustering (HAC)

Partitioning Methods:

- **k-means** (Lloyd's algorithm)
- k-medoids

Spectral Methods:

- Spectral clustering

Other Approaches:

- Community detection
- High-dimensional clustering
- Self-Organizing Maps (SOM)
- Neural Gas

Historical Note: Research on clustering has been active for at least 50 years

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Hierarchical Agglomerative Clustering

Algorithm Principle

Iterative Process:

Step 1: Initialization

- Each data point is a cluster

Step 2: Search

- Find the closest pair of clusters

Step 3: Merge

- Merge these two clusters

Step 4: Iteration

- Repeat from step 2 until completion

Result: Produces a **dendrogram** of the data

Distances Between Sets (Clusters)

Possible Metrics:

- **Distance between centers** (centroid linkage)
- **Maximum distance** (complete linkage): $\max_{x \in C_1, y \in C_2} d(x, y)$
- **Minimum distance** (single linkage): $\min_{x \in C_1, y \in C_2} d(x, y)$

Limitation: The number of clusters is **unknown a priori**

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k-means Strategy

Fundamental Assumptions

The **k-means** clustering algorithm assumes:

- A **normed metric space** (Euclidean) on the data
- An **unsupervised** assignment of data to clusters
- A **hard-assignment** algorithm: assignment is **binary**, each data point is assigned to **one and only one** cluster

Formal Model

Data:

Let $X = \{x_i\}_{i \in [[N]]} \subset \Omega \subseteq \mathbb{R}^D$, and given $K \in \mathbb{N}^*$

Model Variables:

- **Binary assignment variables (latent):**

$$Z = \{z_{ik}\} \quad \text{with} \quad z_{ik} \in \{\text{false}, \text{true}\} \equiv \{0, 1\} \quad \forall i, k$$

- **Cluster representatives:**

$$M = \{\mu_k\}_{k \in [[K]]} \quad \text{with} \quad \mu_k \in \mathbb{R}^D \quad \forall k$$

Complete Parameters:

$$\theta = [M, Z]$$

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k-means Loss Function

Optimization Problem

k-means seeks the following assignment:

$$\hat{\theta} = [\hat{M}, \hat{Z}] = \underset{\theta = [M, Z]}{\arg\min}\, L(\theta, X)$$

Loss Function

Definition:

$$L(\theta, X) = \sum_{i=1}^{N} \sum_{k=1}^{K} z_{ik} \|x_i - \mu_k\|_2^2$$

Interpretation:

- Sum of **squared distances** between each point and its cluster representative
- The smaller this sum, the better the partition

Problem Complexity

Major Difficulty: Since the assignment is binary, exact optimization is **NP-Hard**

Practical Solution: Seek an **approximation via coordinate descent**

Recap: Coordinate Descent Algorithm

General Principle

Coordinate descent is an **alternating minimization algorithm**:

Problem:

Let $f : \mathbb{R}^D \to \mathbb{R}$, we seek:

$$x^* = \arg\min_{x \in \mathbb{R}^D} f(x)$$

Resolution Strategy

Definition of Partial Functions:

For $d \in [[D]]$, define $f^d : \mathbb{R}^D \times \mathbb{R} \to \mathbb{R}$:

$$f^d(x, y) = f([x(1), \dots, x(d-1), y, x(d+1), \dots, x(D)]^T)$$

Alternating Optimization:

Seek alternately the minimum along each coordinate:

$$x^{(t+1)}(d) = \arg\min_{y} f^d(x^{(t)}, y)$$

Principle: Fix all variables except one, optimize this variable, then move to the next

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Application to k-means

Alternating Optimization of Z and M

We **alternate the optimization** of Z and M

Definitions of Partial Losses:

$$L_M(Z) = L(\theta, X)|_{M=M}$$

$$L_Z(M) = L(\theta, X)|_{Z=Z}$$

Optimization with Respect to M (Representatives)

Derivative Calculation:

$$\frac{\partial L_Z}{\partial \mu_k} = -2 \sum_{i=1}^{N} z_{ik}(x_i - \mu_k)$$

Optimality Condition:

The optimal representative M for a given assignment Z is reached when:

$$\frac{\partial L_Z}{\partial \mu_k} = 0 \quad \Rightarrow \quad \mu_k = \frac{\sum_{i=1}^{N} z_{ik} x_i}{\sum_{i=1}^{N} z_{ik}}$$

Fundamental Conclusion: Cluster representatives are their **centers of mass** (barycenters)

Assignment Update

Optimization with Respect to Z

Given a set of cluster representatives M, to minimize $L_M(Z)$:

Recall of the Loss:

$$L_M(Z) = \left[\sum_{i=1}^{N} \sum_{k=1}^{K} z_{ik} \|x_i - \mu_k\|_2^2 \right]_{M=M}$$

Assignment Rule

This is a **sum of positive terms** → assign:

$$z_{ik} = 1 \quad \text{only when} \quad k = \arg\min_m \|x_i - \mu_m\|_2^2$$

(and $z_{ik} = 0$ otherwise)

Conclusion: $L_M(Z)$ is minimized when **each data point is assigned to its nearest cluster representative**

Complete k-means Algorithm

Detailed Description

Input Data:

- $X = \{x_i\}_{i \in [[N]]}$ with $x_i \in \mathbb{R}^D$
- $K \in \mathbb{N}^*$ (number of clusters)

Algorithm Steps

Step 1: Initialization

Initialize cluster representatives $M^{(0)}$

Step 2: Assignment (E-step like)

Given cluster representatives $M^{(t)}$, assignment $Z^{(t)}$ associates each data point with its nearest cluster representative:

$$z_{ik}^{(t)} = \begin{cases} 1 & \text{if } k = \arg\min_m \|x_i - \mu_m^{(t)}\|_2^2 \\ 0 & \text{otherwise} \end{cases}$$

Step 3: Update (M-step like)

Given assignment $Z^{(t)}$, new cluster representatives $M^{(t+1)}$ are the centers of mass of data points assigned to clusters:

$$\mu_k^{(t+1)} = \frac{\sum_{i=1}^{N} z_{ik}^{(t)} x_i}{\sum_{i=1}^{N} z_{ik}^{(t)}}$$

Step 4: Convergence

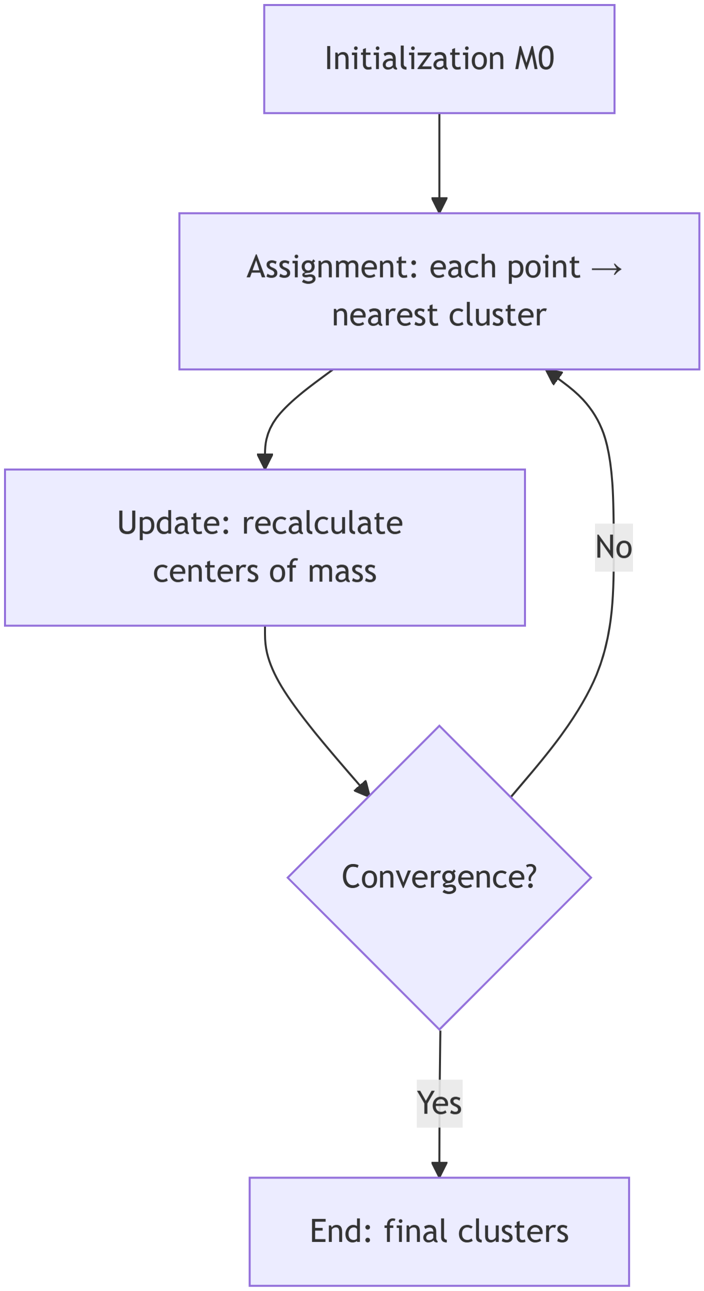
Repeat from step 2 until convergence

Convergence Criteria

Convergence is reached when:

- The centers of mass **do not move much**: $\|\mu_k^{(t+1)} - \mu_k^{(t)}\| < \epsilon$
- Or the assignment is **stable**: $Z^{(t+1)} = Z^{(t)}$

Process Visualization



Important Note: Step 2 is similar to an **Expectation** step, and step 3 is similar to a **Maximization** step, considering a hard assignment (reference to the EM algorithm)

Properties of the k-means Algorithm

Theoretical Guarantees

Loss Decrease:

Since it performs **alternating minimization of convex functions**, it guarantees to **decrease the loss at each iteration**:

$$L(M^{(t+1)}, Z^{(t+1)}) \leq L(M^{(t)}, Z^{(t)})$$

Finiteness:

There is a **combinatorial but finite** number of possible assignments → the number of iterations is **(large but) finite**

Geometric Interpretation

Voronoi Diagram:

Step 2 is equivalent to constructing the **discrete Voronoi diagram** of X with centers M as seeds

Definition: The Voronoi diagram partitions space into regions, where each region contains all points closer to a given center than to any other center

Link with the EM Algorithm

Conceptual Comparison:

- **Step 2 (Assignment):** similar to an **Expectation** step (but hard)
- **Step 3 (Update):** similar to a **Maximization** step

Key Difference: k-means uses **hard assignment** while EM uses **soft assignment**

Sensitivity to Initialization

Problem of Local Optima

Major Limitation: Since exact optimization (optimal assignment) is **NP-Hard**, the k-means algorithm reaches a **local optimum**

Consequences:

- The quality of the result **varies with initialization**
- Different initializations can yield very different results

Practical Strategies

Randomization:

Randomization can be useful if computation is fast → run multiple times with different initializations

Heuristics:

Otherwise, use **heuristics for better initialization** (like k-means++)

Variants, Links, and Relationships

k-means++: Smart Initialization

Fundamental Principle

The principle of **k-means++** is to **maximally disperse** the initial cluster representatives $M^{(0)}$ over the data

Demonstrated Advantages:

1. Produces **higher quality** clusters
2. **Accelerates convergence** of k-means

Detailed k-means++ Algorithm

Step 1: First Center

Select the first cluster representative μ_1^0 **randomly** from the data X

Step 2: Distance Calculation

For all unselected data points x_i, calculate:

$$\Delta_i = \|x_i - \mu_m^0\|_2^2$$

the distance between x_i and its nearest representative μ_m^0 among the k representatives already selected

Step 3: Weighted Sampling

Sample the next representative μ_{k+1}^0 from X with **probability proportional** to the distribution Δ

$$\mathbb{P}(\text{choosing } x_i) = \frac{\Delta_i}{\sum_j \Delta_j}$$

Step 4: Iteration

Repeat from step 2 until K representatives are chosen

Algorithm Intuition

Key Idea: Points **far from already selected centers** have a higher chance of being chosen as new centers

Consequence: Initial centers are well **distributed in the data space**, covering different density regions

Practical Adoption

Widespread Use: This initialization strategy is used by several Data Science packages:

- MATLAB™
- Python SciKit Learn
- R
- And other standard libraries

Relationship Between k-means and Gaussian Models

k-means as a Special Case of GMM

Question: Why do we say that k-means considers a **Gaussian model** for clusters?

Answer:

k-means can be seen as a limiting case of GMM (Gaussian Mixture Model) where:

Model Assumptions:

- Each cluster follows a **normal distribution**
- **Spherical and identical** covariance matrices: $\Sigma_k = \sigma^2 I_D$ for all clusters
- Variance $\sigma^2 \to 0$: distributions become spikes

Consequence:

In this limit, the **soft assignment** of GMM becomes a **hard assignment**, and we recover k-means

Probabilistic Formulation

If we consider the probabilistic model:

$$\mathbb{P}(x|\mu_k, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{D/2}} \exp\left(-\frac{\|x-\mu_k\|^2}{2\sigma^2}\right)$$

When $\sigma \to 0$, maximizing this probability is equivalent to minimizing $\|x-\mu_k\|^2$, which is exactly the k-means criterion

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Intrinsic Link with the Euclidean Metric

Why k-means Depends on the Euclidean Metric?

Analysis of the Loss Function:

$$L(\theta, X) = \sum_{i=1}^{N}\sum_{k=1}^{K} z_{ik}\|x_i - \mu_k\|_2^2$$

The norm $\|\cdot\|_2$ is the **Euclidean norm**: $\|x\|_2^2 = \sum_d x(d)^2$

Practical Consequences

Scale Sensitivity:

k-means is **sensitive to scale**. Variables with large values dominate the distance

Solution: Normalize the data before applying k-means:

- Center: subtract the mean
- Scale: divide by the standard deviation

Limitation to Euclidean Spaces:

k-means cannot be directly applied to **non-Euclidean** data (graphs, texts, etc.)

Alternatives: For other metrics, use variants like k-medoids that work with arbitrary distances

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Constrained Clustering

k-means Extension

Clustering can be **constrained** with:

MUST-LINK Constraints:

Two points must be in the **same cluster**

CANNOT-LINK Constraints:

Two points must be in **different clusters**

Practical Application

These constraints allow incorporating **prior knowledge** or **domain constraints** into the clustering process

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Synthesis of Key Concepts

Essential Characteristics of k-means

Nature of the Method:

- k-means is part of the **unsupervised** family of data modeling techniques
- One of the **most popular** clustering techniques

Assignment:

- k-means performs **hard assignment**
- Each point belongs to **one cluster only**

Optimization:

- **Solid** but **NP-Hard** optimization criterion (loss)
- The k-means algorithm seeks an **approximate solution** via coordinate descent (alternating minimization)

Convergence:

- The loss **is not convex** → technique **sensitive to initialization**
- k-means++ is a **prior heuristic** for initialization, effective in practice

Extensions:

- Clustering can be **constrained** (MUST-LINK and CANNOT-LINK constraints)

Comparison: k-means vs GMM vs Hierarchical Clustering

Criterion	k-means	GMM (with EM)	Hierarchical Clustering
Assignment Type	Hard	Soft	Hierarchical
Cluster Shape	Spherical(same variance)	Elliptical(different covariances)	Flexible
Number of Clusters	Must be specified	Must be specified	Determined a posteriori(dendrogram)

Criterion	k-means	GMM (with EM)	Hierarchical Clustering
Complexity	$O(N \cdot K \cdot D \cdot T)$where T is the number of iterations	$O(N \cdot K \cdot D^2 \cdot T)$	$O(N^2 \log N)$ or $O(N^3)$
Advantages	Fastsimple	Probabilistic modelvaried cluster shapes	No need to specify Kvisualization via dendrogram
Disadvantages	Sensitive to initializationassumes spherical clusters	Slowermore parameters to estimate	Expensive for large datano reassignment

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Choosing the Number of Clusters K

Problem

Difficulty: The number of clusters K must be **specified a priori** in k-means

Selection Methods

Elbow Method:

Plot the loss L as a function of K and look for an "elbow" in the curve

Silhouette Score:

Measures clustering quality by comparing intra-cluster distance to inter-cluster distance

Gap Statistic:

Compares observed loss to expected loss under a reference distribution

Information Criteria:

- BIC (Bayesian Information Criterion)
- AIC (Akaike Information Criterion)

Practical Advice: Test several values of K and use multiple criteria

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Typical Exam Questions

Conceptual Questions

Formal Description:

- Formally describe clustering
- In what sense is it an unsupervised technique?

Relationship with Metric:

- Explain why k-means is intrinsically linked to the Euclidean metric
- Why do we say that k-means considers a Gaussian model for clusters?

Accuracy:

- Is the k-means algorithm exact?
- What are the theoretical guarantees?

Technical Questions

Initialization:

- What are the principles for initializing k-means?
- Describe the k-means++ algorithm

Optimization:

- What is the coordinate descent algorithm?
- How does it apply to k-means?

Relationship with Other Methods:

- What is the relationship between k-means and GMM?
- What is the difference between hard and soft assignment?

Mathematical Developments

Important Advice: It is **strongly recommended** to develop the algebra contained in this chapter

Key Points to Master:

- Derivation of the loss function
- Calculation of $\frac{\partial L_Z}{\partial \mu_k}$ and optimality condition
- Proof that centers are barycenters
- Proof of monotonic loss decrease
- Relationship between k-means and limiting Gaussian model

Mathematical Concepts to Master

Algebra and Geometry

Fundamentals:

- Euclidean norm and distance
- Center of mass (barycenter)
- Voronoi diagram
- Metric spaces

Optimization

Techniques:

- Coordinate descent
- Alternating minimization
- Combinatorial optimization
- NP-Hard complexity

Properties:

- Convergence of iterative algorithms
- Local vs global optima
- Convex functions

Probability and Statistics

Concepts:

- Multivariate Gaussian distribution
- Maximum likelihood
- Mixture models

Practical Implementation Tips

Data Preprocessing

Recommended Steps:

1. **Normalization:** center and scale variables
2. **Outlier Management:** detect and handle outliers
3. **Dimensionality Reduction:** PCA if D is very large

Algorithmic Parameters

Initialization:

- Use **k-means++** by default
- Or run **multiple times** with random initializations

Convergence:

- Tolerance threshold: typically $\epsilon = 10^{-4}$
- Maximum number of iterations: avoid infinite loops

Choice of K:

- Test several values
- Use validation criteria